

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

MindMosaic

Software Engineering Intern

Aug 2025 – May 2026

East Lansing, MI

- Built and shipped an interactive knowledge-graph platform that keeps dense, interconnected content navigable as it grows, delivering a production MVP to real users on a polyglot Spring Boot, Neo4j, and PostgreSQL stack.
- Migrated hot read paths off recursive SQL self-joins onto a Neo4j graph layer so relationships stay first-class, cutting query latency 48% (850ms to 440ms) and making multi-hop traversals 3x faster than the equivalent SQL.

Secure and Efficient Autonomous Systems Lab

Undergraduate Research Assistant

May 2026 – Present

East Lansing, MI

- Built the full language-to-motion pipeline in simulation on ROS2 and Gazebo with the attack layer decoupled at the ROS2 interface, so swapping the target robot platform cost only the URDF and control layer and never touched the security result.
- Researching whether the cheap triage layer that guards an LLM-controlled robot is itself a gameable security boundary, by splitting one unsafe command into individually innocent fragments that reassemble past the gate into unsafe robot behavior.

Robostangs (FRC Team 548)

Software Captain

Jun 2023 – May 2024

Northville, MI

- Led the software and electrical teams (5 and 6 engineers) for a competition robot that reached the FIRST World Championship and placed top 16 in its division in Houston, the program's strongest world-stage finish since 2014.
- Anchored auto-aim on real-time, vision-corrected localization that fused two AprilTag cameras with swerve odometry on a dedicated 250Hz thread with latency compensation, holding pose accurate during aggressive motion.

Claude Builder Club @ MSU

Co-Founder & Vice President

Oct 2025 – Present

East Lansing, MI

- Built event check-in as a single atomic database RPC that validates a QR token, records attendance, and awards points server-side, making door-time check-ins concurrency-safe and idempotent so simultaneous scans cannot double-award or be forged client-side, sustaining 150+ check-ins per event.

PROJECTS

Dadei | *Python, FastAPI, Redis, PostgreSQL, Docker*

dadei.app

Voice assistant for human-approved calendar, email, and task actions

- Solved the same-room, multiple-microphone problem with two arbitration layers, a live utterance-owner election and a 400ms windowed quality election scoring SNR, clarity, and speaker cadence, so several devices hearing one utterance transcribe it once via the best mic and cut duplicate transcription roughly in half.

fliks | *Go, PostgreSQL, AWS EC2*

github.com/fliks-gg

Crowd-validated gameplay skill platform with shareable certification cards

- Chose a single Go monolith over a microservice split, keeping auth, rate-limiting, and rating aggregation in-process because premature decomposition is the harder call to defend at pre-scale, one deploy unit against one datastore.

TECHNICAL SKILLS

Languages Python, Java, Go, C++, JavaScript/TypeScript, SQL

Frameworks Spring Boot, FastAPI

Cloud & Infrastructure AWS (EC2, S3, RDS), CI/CD

Robotics ROS2, Gazebo, AprilTag

Databases PostgreSQL, Neo4j, Redis